

Session 7: Hands-on data analysis using R (May 19, 2026, 2pm-3:30pm ET)

Overview of the session:

We will cover an overview of NHANES data and considerations for analysis, how to import and merge the NHANES oral microbiome data, demographic data, and dietary data in R, analysis of relationships between demographic and lifestyle factors with the oral microbiome (alpha diversity, beta diversity, taxa composition at the genus level), and how to create figures/tables to display results that are commonly seen in the microbiome field

Please make sure to do the following **prior to the session:**

1. **Download R and R studio**
 - a. R and R studio are 2 separate downloads
 - b. Walk through section A.1 through A.4 here: <https://rstudio-education.github.io/hopr/starting.html>
2. **Download and load the following R packages**
 - a. tidyverse, ggplot2, vegan, survey, haven, gtsummary

Example code for downloading and loading packages:

```
#Download packages (you only need to do this once)
install.packages("tidyverse")
install.packages("ggplot2")
install.packages("vegan")

#Load packages (you will need to run this code each time
#you open R and want to use the packages)
library(tidyverse)
library(ggplot2)
library(vegan)
```

3. **Download the following NHANES data** *(note: For the oral microbiome NHANES data, some of the microbiome data files are really large (beta diversity and abundance) and may take a few minutes to download.)*

NHANES data is organized in cycles. Each cycle is 2 years. The oral microbiome data was collected from 2009-2012 (cycles 2009-2010 [F], 2011-2012 [G]); both cycles are in the same data files. When downloading demographic, dietary, and questionnaire data, make sure to download from both cycles.

Demographic, dietary, and questionnaire data by cycle can be accessed here:

<https://wwwn.cdc.gov/nchs/nhanes/default.aspx>

Files to download:

- a. Demographics data:
 - File name: Demographic Variables & Sample Weights
 - o Data file: DEMO_F Data
 - o Data file: DEMO_G Data
- b. Dietary data:
 - File name: Dietary Interview – Total Nutrient Intakes, First Day
 - o Data file: DR1TOT_F Data
 - o Data file: DR1TOT_G Data
- c. Oral microbiome data: <https://wwwn.cdc.gov/Nchs/Nhanes/omp/>
 - Alpha diversity
 - o Data file: dada2rsv-alpha.txt
 - Beta diversity

- Data file: dada2rsv-braycurtis-beta.txt

Abundance

- Data file: dada2rsv-phylum-relative.txt
- Data file: dada2rsv-genus-relative.txt
- Data file: dada2rsv-phylum-count.txt
- Data file: dada2rsv-taxonomy-annotate.txt
- Data file: dada2rsv-genus-count.txt